

Aircrack-ng: Securing Wireless Networks

Aircrack-ng is not a single tool, but rather a suite of tools that can be used to hack a wireless network. In this article, though, it is used to secure a wireless network by discovering its vulnerabilities. Read on to see how this can be done.

Security in networks is very vital for small as well as large organisations. Not only does it help in maintaining the confidentiality of a client's and employer's data, but it is also important for retaining trade secrets to overcome competition. Wireless networks form an important mode of communication as wired networks tend to incur more infrastructure costs. But are wireless networks sufficiently secure? Let's have a look at a common scenario.

Bob's start-up company, XYZ Pvt Ltd, has a few wireless access points. He has his network configured over the WEP protocol with simple dictionary based passwords like 'YOU CANT GUESS'. He is really annoyed with the repeated 'unauthentication' of legitimate users from the Wi-Fi network. After a few such compromises, he changes his Wi-Fi passwords but the attacks continue. This continuous mischief irritates him and he has no idea why this is happening.

He contacts Eve, a security analyst, for help. Eve investigates the situation and explains how wireless protocols function. She provides him with the basic details of the two algorithms used in a majority

of wireless networks: WEP (Wired Equivalent Privacy) and WPA/WPA2 (Wi-Fi Protected Access).

Wired Equivalent Privacy (WEP)

This algorithm is based on the RC4 stream cipher and CRC checksum mechanism to provide confidentiality and integrity. Open System authentication and Shared Key authentication are the two methods of authentication used in WEP.

1. *Open System authentication:* In this case, the WLAN client need not provide its credentials to the access point during authentication. Any client can authenticate with the access point.
2. *Shared Key authentication:* In this authentication mechanism as shown in figure 1, the WEP key is used for authentication in a four-step handshake process:
 - a. The client sends an authentication request to the access point.
 - b. The access point responds to the request with a clear-text challenge.

